STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 4: Laws of Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

Mass.

Suggested allocation by section: 1-mark questions earn one mark for the correct fact, formula or unit; 2-mark questions normally award one mark for the principle/formula and one for the explanation or result; 3-mark questions award credit for the relation, working/reasoning and result with unit; 4-mark questions award credit for the principle, equation, working and final answer/interpretation. Accept equivalent correct methods.

Q2 1 mark

sumF=ma.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q3 1 mark

No.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q4 1 mark

p=mv.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q5 1 mark

Zero.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q6 1 mark

Opposes relative sliding.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q7 1 mark

Inward resultant of actual forces.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q8 1 mark

muN.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q9 2 marks

4 m/s^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q10 2 marks

-3 kg m/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q11 2 marks

98 N.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q12 2 marks

24 N.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q13 2 marks

10 N s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q14 2 marks

Water exerts a forward reaction.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q15 3 marks

4 m/s^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q16 3 marks

-60 N.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q17 3 marks

5 m/s^2; 86.6 N.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q18 3 marks

20 N inward.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q19 4 marks

a=5 m/s^2; v=15 m/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q20 4 marks

Deltap=-10 N s; force=-100 N.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.